

[ACCELERATE] Accelerate

Accelerate documentation Accelerate and get access to get started Accelerate is a library that enables any distributed ...

来源：<https://huggingface.co/docs/accelerate/index>

分类：12_模型部署与工程化

Accelerate documentation

Accelerate

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to get started

Accelerate is a library that enables the same PyT configuration by adding just four lines of code! made simple, efficient and adaptable.

```
+ from accelerate import Accelerator
```

```
+ accelerator = Accelerator()
```

```
+ model, optimizer, training_dataloader, scheduler
```

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```

```
+ )
```

```
for batch in training_dataloader:
```

```
optimizer.zero_grad()
```

```
inputs, targets = batch
```

```
inputs = inputs.to(device)
```

```
targets = targets.to(device)
```

```
outputs = model(inputs)
```

```
loss = loss_function(outputs, targets)
```

```
+ accelerator.backward(loss)
```

```
optimizer.step()
```

```
scheduler.step()
```

Built on `torch_xla` and `torch.distributed`, Accelerate don't have to write any custom code to adapt to the hardware. Convert existing codebases to utilize DeepSpeed, have automatic support for mixed-precision training.

`torch_xla`

`torch.distributed`

To get a better idea of this process, make sure to read the documentation.

This code can then be launched on any system through:

```
accelerate launch {my_script.py}
```

Learn the basics and become familiar with using Accelerate for the first time!

Practical guides to help you achieve a specific goal, how to use Accelerate to solve real-world problems.

High-level explanations for building a better understanding, avoiding subtle nuances and pitfalls in distributed training.

Technical descriptions of how Accelerate classes work.